



Question 1) After passing through a Stern-Gerlach magnet oriented along an arbitrary axis $\mathbf{n} = (\sin \beta \cos \alpha, \sin \beta \sin \alpha, \cos \beta)$, the beam is split into two spin states:

$$\begin{aligned} |S_{\mathbf{n}}; +\rangle &= \cos \frac{\beta}{2} |+\rangle + e^{i\alpha} \sin \frac{\beta}{2} |-\rangle, \\ |S_{\mathbf{n}}; -\rangle &= -e^{-i\alpha} \sin \frac{\beta}{2} |+\rangle + \cos \frac{\beta}{2} |-\rangle. \end{aligned} \quad (1)$$

a) Find these states by diagonalizing the spin operator

$$\begin{aligned} \hat{S}_{\mathbf{n}} &= n_x \hat{S}_x + n_y \hat{S}_y + n_z \hat{S}_z \\ &= \frac{\hbar}{2} (\cos \beta |+\rangle \langle +| + e^{-i\alpha} \sin \beta |+\rangle \langle -| + e^{i\alpha} \sin \beta |-\rangle \langle +| - \cos \beta |-\rangle \langle -|) \end{aligned} \quad (2)$$

$\hat{S}_{\mathbf{n}}$ along the axis $\mathbf{n}$. That is, treat this as an eigenvalue problem (similarly to what S&N tells us to do in **Problem 1.11**). Feel free to use the matrix representation for our states

$$|+\rangle = \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \quad |-\rangle = \begin{pmatrix} 0 \\ 1 \end{pmatrix} \quad (3)$$

and whatever method you prefer.

b) Calculate the inner product $\langle S_{\mathbf{n}}; + | \hat{S}_{\mathbf{n}} | + \rangle$. Are the states $|S_{\mathbf{n}}; +\rangle$ and $\hat{S}_{\mathbf{n}} |+\rangle$ the same?

c) In what direction does the Bloch vector of a Stern-Gerlach magnet point for the states $|S_{\mathbf{n}}; +\rangle$ and $|S_{\mathbf{n}}; -\rangle$? Recall that the Bloch vector for a state $|\psi\rangle$ is given by $\langle \hat{\mathbf{S}} \rangle = \langle \psi | \hat{\mathbf{S}} | \psi \rangle$.

d) Each Stern-Gerlach magnet can be described by a projector operator

$$\hat{P}_{\mathbf{n}; \pm} = |S_{\mathbf{n}}; \pm\rangle \langle S_{\mathbf{n}}; \pm|. \quad (4)$$

After passing through a magnet oriented in the direction $\mathbf{n}$ that blocks $\mp$, but allows $\pm$, we have a pure state

$$|S_{\mathbf{n}}; \pm\rangle. \quad (5)$$

We do not ask about what happened earlier than this point, all we care about is that this new state was created. Suppose now we add a second magnet oriented in the direction $\mathbf{m}$. After exiting through the second magnet, the probability of finding the particle in the state $|S_{\mathbf{m}}; \pm\rangle$ is given by

$$\langle S_{\mathbf{n}}; \pm | \hat{P}_{\mathbf{m}; \pm} | S_{\mathbf{n}}; \pm \rangle = |\langle S_{\mathbf{m}}; \pm | S_{\mathbf{n}}; \pm \rangle|^2. \quad (6)$$

You can think of this as the probability or the relative intensity of the beam. Imagine we add a third magnet oriented in the direction $\mathbf{l}$. What is the probability of finding the particle in the state $|S_{\mathbf{l}}; \pm\rangle$ after passing through all three magnets? Express your answer in terms of the angle $\theta_{\mathbf{lm}}$ between the directions $\mathbf{l}$ and $\mathbf{n}$ and the angle $\theta_{\mathbf{mn}}$ between the directions $\mathbf{m}$ and $\mathbf{n}$.

Hint: you may want to start by proving that the projector $\hat{P}_{\mathbf{n}; \pm}$ is given by

$$\hat{P}_{\mathbf{n}; \pm} = \frac{1 + \mathbf{n} \cdot \hat{\mathbf{S}} / |\mathbf{S}|}{2} \quad (7)$$

e) Calculate the probability of finding the particle in the state $|S_{\mathbf{l}}; \pm\rangle$ after passing through all three magnets for the following arrangements:

$$(a) \quad \mathbf{n} = \hat{\mathbf{z}}, \quad \mathbf{m} = -\hat{\mathbf{z}}, \quad \mathbf{l} = \hat{\mathbf{z}}$$

$$(b) \quad \mathbf{n} = \hat{\mathbf{z}}, \quad \mathbf{m} = \hat{\mathbf{x}}, \quad \mathbf{l} = \hat{\mathbf{z}}$$

$$(c) \quad \mathbf{n} = \hat{\mathbf{z}}, \quad \mathbf{m} = (\hat{\mathbf{x}} + \hat{\mathbf{z}})/\sqrt{2}, \quad \mathbf{l} = \hat{\mathbf{x}}$$

- g) Generalize your result for an arbitrary number N of sequential Stern-Gerlach magnets going from an initial $\hat{\mathbf{z}}$ direction to a final $\hat{\mathbf{x}}$ direction in equal angular steps confined to the xz -plane. What happens in the limit $N \rightarrow \infty$? What is the probability of measuring the particle in the state $|S_{\mathbf{x}}; \pm\rangle$? How about $|S_{\mathbf{z}}; \pm\rangle$? How does this compare with case (b) above?